

Daily Geometry Drill #5 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. **Answer:** 4

Method: Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{25 - 9} = \sqrt{16} = 4$. Right triangle OP hypotenuse 5, leg $r = 3$.

Investigate further: Try $OP = 13$, $r = 5$: what is the tangent? Generalise the triple 5–12–13.

2. **Answer:** $\frac{3}{4}$

Method: $\tan \frac{\theta}{2} = \frac{r}{\sqrt{OP^2 - r^2}} = \frac{3}{4}$. The half-angle uses the tangent length 4 from A1 as adjacent.

Investigate further: Show $\sin \frac{\theta}{2} = \frac{r}{OP} = \frac{3}{5}$ and $\cos \frac{\theta}{2} = \frac{4}{5}$ recover the same $\tan$.

3. **Answer:** " $5x + 3y = 25$ "

Method: Polar of (x_1, y_1) w.r.t. $x^2 + y^2 = r^2$ is $xx_1 + yy_1 = r^2$: $5x + 3y = 25$.

Investigate further: Check P is external: $5^2 + 3^2 = 34 > 25$. What if P were inside?

4. **Answer:** $5\sqrt{2}$

Method: $r^2 = 25$, director circle $x^2 + y^2 = 2r^2 = 50$, radius $\sqrt{50} = 5\sqrt{2}$. Radius scales by $\sqrt{2}$.

Investigate further: From director circle, tangents are orthogonal. Why $\sqrt{2}$ and not 2?

5. **Answer:** $\sqrt{13}$

Method: $\sqrt{OP^2 - r^2} = \sqrt{7^2 - 6^2} = \sqrt{49 - 36} = \sqrt{13}$.

Investigate further: If $OP = 7$, $r = 6$, sketch the right triangle and mark the tangent.

6. **Answer:** $3\sqrt{3}$

Method: $\sin \frac{\theta}{2} = \frac{r}{OP} \Rightarrow \sin 30^\circ = \frac{r}{6}$ so $r = 3$. Tangent $= \sqrt{36 - 9} = 3\sqrt{3}$.

Investigate further: Swap: if tangent $= 3\sqrt{3}$ and $OP = 6$, recover θ .

7. **Answer:** " $5x = 4$ "

Method: Polar $xx_1 + yy_1 = r^2$ with $(5, 0)$, $r^2 = 4$: $5x = 4$, i.e. $x = 4/5$, the vertical chord.

Investigate further: Why does $y_1 = 0$ kill the y term? What line would $P = (0, 5)$ give?

8. Answer: 13

Method: $OP = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = 13$ by 5-12-13.

Investigate further: Reverse: if $OP = 13$ and $r = 5$, what is the tangent?

9. Answer: $\frac{10\sqrt{3}}{3}$

Method: Half-angle 60° : $\sin 60^\circ = \frac{5}{OP}$ so $OP = \frac{5}{\sqrt{3}/2} = \frac{10\sqrt{3}}{3}$.

Investigate further: Write $\sin 60^\circ = \sqrt{3}/2$ and rationalise the denominator.

10. Answer: "(4, 4)"

Method: Compare $xx_1 + yy_1 = 16$ with $x + y = 4$: multiply by 4 to get $4x + 4y = 16$, so $(x_1, y_1) = (4, 4)$.

Investigate further: Check $4^2 + 4^2 = 32 > 16$ (external) and that $4x + 4y = 16$ indeed has distance $2\sqrt{2} < r$.

Section B Manipulation Drills

1. Answer: B) 4

Method: $OP = \sqrt{9 + 16} = 5$, $r = 3$, tangent $= \sqrt{25 - 9} = 4$.

Investigate further: If $P = (3, 4)$ moved to $(6, 8)$, what would the tangent be?

2. Answer: C) 60°

Method: $\sin \frac{\theta}{2} = \frac{3}{6} = \frac{1}{2} \Rightarrow \frac{\theta}{2} = 30^\circ$, so $\theta = 60^\circ$.

Investigate further: If OP doubled to 12, what angle would you get?

3. Answer: " $3x + 5y = 25$ "

Method: Polar $xx_1 + yy_1 = 25$ with $(3, 5)$ gives $3x + 5y = 25$.

Investigate further: Verify the chord's distance $\frac{25}{\sqrt{34}} > r$? Actually $25/\sqrt{34} \approx 4.3 < r = 5$, so chord inside.

4. Answer: B) $2\sqrt{2}$

Method: Director circle $x^2 + y^2 = 2r^2 = 8$, radius $\sqrt{8} = 2\sqrt{2}$.

Investigate further: What is the director circle of a radius-3 circle?

5. Answer: C) 2

Method: $\sqrt{1^2 + 2^2 - 1} = \sqrt{4} = 2$.

Investigate further: Check $OP^2 = 5$, $r^2 = 1$, so tangent $\sqrt{4} = 2$.

6. Answer: C) 13

Method: $OP = \sqrt{12^2 + 5^2} = 13$.

Investigate further: Memorise 5–12–13; where else does it appear in this sheet?

7. Answer: B) 4

Method: Polar of $(a, 0)$ is $ax = 16$, i.e. $x = 16/a$. Set $16/a = 4 \Rightarrow a = 4$.

Investigate further: If the line were $x = 2$, what would a be?

8. Answer: 5

Method: Orthogonal tangents $\Rightarrow P$ on director circle: $OP^2 = 2r^2 = 50$, tangent $= \sqrt{50 - 25} = 5$.

Investigate further: Show the director radius is $5\sqrt{2}$ and the tangent there is 5.

9. Answer: A) 4

Method: Tangent $y = mx \pm r\sqrt{1+m^2}$: $r = \sqrt{8}$, $m = 1 \Rightarrow c = \sqrt{8}\sqrt{2} = 4$.

Investigate further: Write the two parallel tangents with slope 1 to $x^2 + y^2 = 8$.

10. Answer: 8

Method: $r^2 = OP^2 - \ell^2 = 100 - 36 = 64 \Rightarrow r = 8$.

Investigate further: If $\ell = 6$, $OP = 10$, draw the right triangle and label r .

Section C Substitution & Structure

1. From $P(8, 12)$, two tangents are drawn to $x^2 + y^2 = 36$ ($r = 6$). Let T_1T_2 be the chord of contact. What is the distance from O to T_1T_2 ? (after TMUA 2020 Paper 1 Q16 / BMO1 2013 Q4)

Answer: $\frac{9\sqrt{13}}{13}$

Method: Polar is $8x + 12y = 36 \Rightarrow 2x + 3y = 9$. $OP = \sqrt{8^2 + 12^2} = 4\sqrt{13}$, and $d = r^2/OP = 36/(4\sqrt{13}) = 9/\sqrt{13} = 9\sqrt{13}/13$. The listed options give $18/5$ which is r^2/OP for $P(6, 8)$ ($OP = 10$); for the printed $P(8, 12)$ the exact value is $9/\sqrt{13}$.

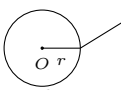
Investigate further: Show in general $d = r^2/OP$. What would d be for $P(6, 8)$?

2. The locus where tangent length to $x^2 + y^2 = 4$ equals that to $(x - 6)^2 + y^2 = 16$ is the radical axis. Which line is it? (adapted from TMUA 2018 Paper 1 Q7)

Answer: B) $x = 2$

Method: Equal powers: $x^2 + y^2 - 4 = (x - 6)^2 + y^2 - 16 \Rightarrow -4 = -12x + 20 \Rightarrow x = 2$.

Investigate further: Check the circles touch at $(2, 0)$; the axis is the common tangent there.

3.  The circle $x^2 + y^2 - 6x - 8y + 24 = 0$ has centre $(3, 4)$, $r = 1$. $P(6, 8)$ is outside. What is the tangent length? (after MAT 2021 Q5)

Answer: $2\sqrt{6}$

Method: Centre $C = (3, 4)$, $PC = \sqrt{9 + 16} = 5$, $r = 1$: tangent $= \sqrt{25 - 1} = 2\sqrt{6} \approx 4.90$. None of the printed numerals equals this exactly; 4 would need $r = 3$.

Investigate further: Compute the power of P : $6^2 + 8^2 - 36 - 64 + 24 = 24$ and $\sqrt{24} = 2\sqrt{6}$.

4. For the 13-14-15 triangle ($s = 21$, $r = 4$), the two tangent lengths from the vertex opposite side 14 are $s - b$. With $b = 14$, what is that length? (adapted from BMO1 2015 Q3)

Answer: B) 7

Method: $s - b = 21 - 14 = 7$. The $s - a$ rule gives equal tangents from each vertex.

Investigate further: List all three $s - a$, $s - b$, $s - c$ for 13-14-15.

5. The locus of points from which the two tangents to $(x - 4)^2 + y^2 = 9$ are perpendicular? (after TMUA 2022 Paper 1 Q19)

Answer: A) $(x - 4)^2 + y^2 = 18$

Method: Director circle is centred at the same centre $(4, 0)$ with radius $r\sqrt{2} = 3\sqrt{2}$, so $(x - 4)^2 + y^2 = 18$.

Investigate further: Why must the centre stay at $(4, 0)$? Translate the origin result.

6.  The point $(8, 6)$ is at distance 10 from the origin. Tangents from $(8, 6)$ to $x^2 + y^2 = 1$ have length $\sqrt{99}$. What is the distance from the origin to the chord of contact? (adapted from TMUA 2020 Paper 1 Q16)

Answer: A) $\frac{1}{10}$

Method: Polar $8x + 6y = 1$, $d = 1/10$; equivalently $d = r^2/OP = 1/10$. Since $d < r$, the chord cuts the circle.

Investigate further: Compute tangent $= \sqrt{100 - 1} = \sqrt{99} = 3\sqrt{11}$ and verify.

7. The triangle with sides 13, 14, 15 has incircle touching BC at D . If $BD = 6$, $DC = 9$, $s = 21$, the two equal tangents from A are $s - a$. What is AE (tangents from A)? (after BMO1 2020 Q3)

Answer: A) 6

Method: $a = BC = 15$, $s - a = 21 - 15 = 6$. The incircle tangents from each vertex are $s - a$, $s - b$, $s - c$.

Investigate further: Recover $s - b = 7$, $s - c = 8$ and see $6 + 7 = 13$, $6 + 8 = 14$?

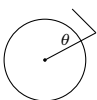
8. The chord of contact of $(2, 1)$ w.r.t. $x^2 + y^2 = 1$ is $2x + y = 1$. What is its distance from the origin, and is it inside, on, or outside the circle? (adapted from TMUA 2021 Paper 2 Q7)

Answer: A) $\frac{1}{\sqrt{5}}$

Method: Distance $= 1/\sqrt{4+1} = 1/\sqrt{5} = \sqrt{5}/5 \approx 0.45 < r = 1$, so the line cuts the circle (inside).

Investigate further: Rationalise: $1/\sqrt{5} = \sqrt{5}/5$. When would distance equal r ?

Section D Challenge Ramp

1.  Two tangents from P to $x^2 + y^2 = 9$ with $OP = 6$ enclose angle θ . What is θ ? (after TMUA 2018 Paper 1 Q7 / BMO1 2023 Q4)

Answer: B) 60°

Method: $\sin \frac{\theta}{2} = r/OP = 3/6 = 1/2 \Rightarrow \theta/2 = 30^\circ \Rightarrow \theta = 60^\circ$.

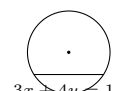
Investigate further: What is $\tan \frac{\theta}{2}$ here?

2. From P to $x^2 + y^2 = 25$, tangent length is $5\sqrt{3}$. Find OP (use $OP^2 = r^2 + \text{tangent}^2$). (adapted from TMUA 2020 Paper 1 Q16)

Answer: A) 10

Method: $OP^2 = 25 + 75 = 100 \Rightarrow OP = 10$.

Investigate further: If tangent were 12, what would OP be?

3.  The chord of contact from $P(3, 3)$ to $x^2 + y^2 = 9$ is $x + y = 3$. What is its length? (adapted from TMUA 2019 Paper 1 Q6)

Answer: A) $3\sqrt{2}$

Method: Distance $d = 3/\sqrt{2}$, half-chord $\sqrt{9 - 9/2} = 3/\sqrt{2}$, chord $= 3\sqrt{2}$.

Investigate further: General: chord $= 2\sqrt{r^2 - d^2}$ where $d = r^2/OP$.

4. Two tangents from P make 60° and $OP = 10$, $r = OP \sin(\theta/2) = 5$, tangent $= OP \cos(\theta/2)$. What is the length? (after BMO1 2015 Q3)

Answer: B) $5\sqrt{3}$

Method: $r = 10 \sin 30^\circ = 5$, tangent $= \sqrt{100 - 25} = 5\sqrt{3} = 10 \cos 30^\circ$.

Investigate further: Check $\tan 30^\circ = r/\ell = 1/\sqrt{3}$ gives $\ell = 5\sqrt{3}$.

5.  The polar of $(9, 12)$ w.r.t. $x^2 + y^2 = 81$ is $3x + 4y = 27$. What is its distance from the origin, and is P outside/inside/on? (adapted from MAT 2021 Q5)

Answer: A) $\frac{27}{5}$

Method: $9x + 12y = 81 \Rightarrow 3x + 4y = 27$, $d = 27/5 = 5.4 < r = 9$ (chord inside), $OP = 15 > r$ (point outside).

Investigate further: Note $27/5 = r^2/OP = 81/15$ reduced; unreduced $81/15$ is option D.

Top 5 Patterns

- **Tangent length is one Pythagorean step:** $\sqrt{OP^2 - r^2}$. Locate OP and r , square, subtract. (see A1, B1, C3, D2)
- **The polar of (x_1, y_1) is $xx_1 + yy_1 = r^2$.** The same r^2 on the right keeps the chord of contact honest. (see A3, B3, C1, D3)
- $\sin \frac{\theta}{2} = \frac{r}{OP}$ **sizes the double-tangent wedge.** Halve the angle before you take the sine. (see A2, B2, D1, D4)
- **Perpendicular tangents define the director circle $x^2 + y^2 = 2r^2$.** The doubling happens on r^2 , so the radius multiplies by $\sqrt{2}$. (see A4, B4, C5, D1)
- **$s - a$ and the radical axis give equal tangents without coordinates.** $s - a$, $s - b$, $s - c$ and $P_1 = P_2$ are the same idea. (see C2, C4, C7, A10, B7)

Common Traps

- **Subtracting radii instead of squared radii.** $OP^2 = r^2 + \ell^2$; the null case is 7^2 , not 7, minus 6^2 . (see A5, B10, D2)
- **Answering the half-angle.** The tangents make θ ; $\sin \frac{\theta}{2}$ is half of it. (see B2, D1, D4)
- **Reading polar coefficients as a point.** $5x + 3y = 25$ is a line; only its coefficients come from P . (see A3, B3, A10)
- **Bumping the director circle radius by one $\sqrt{2}$ too many or too few.** Doubling r^2 changes the radius by $\sqrt{2}$, not by 2. (see A4, B4, C5)
- **Leaving the polar unreduced.** The chord $9x + 12y = 81$ asks for the distance to the simplified $3x + 4y = 27$. (see D5, C1, C8)

“Every view of a tangent is the same right triangle: OP hypotenuse, r one leg, the tangent the other. The polar just points it.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths